

Experiment # 09

Configuring Dynamic Routing Protocol: Routing Information Protocol (RIP)

Objective:

This Lab explains how to configure Routing Information Protocol using commands and their parameters in detail with examples.

Software Tools:

- Cisco Packet Tracer

Theory:

Dynamic Routing Protocol: Routing Information Protocol (RIP)

There are three basic types of routing protocols.

- 1. Distance-vector Routing Protocols:** It uses simple algorithms that calculate a cumulative distance value between routers based on hop count.
Example: Routing Information Protocol Version 1 (RIPv1) and Interior Gateway Routing Protocol (IGRP)
- 2. Link-state Routing Protocols:** Link-state Routing Protocols use sophisticated algorithms that maintain a complex database of internetwork topology.
Example: Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS)
- 3. Hybrid Routing Protocols:** Hybrid Routing Protocols use a combination of distance-vector and link-state methods that tries to incorporate the advantages of both and minimize their disadvantages.
Example: Enhanced Interior Gateway Routing Protocol (EIGRP), Routing Information Protocol Version 2 (RIPv2)

All the above are categorized as IGP & EGP protocols.

Interior Gateway Protocol (IGP) is a Routing Protocol which is used to find network path information within an Autonomous System.

Known Interior Gateway Protocol (IGP) Routing Protocols are Routing Information Protocol (RIP), Interior Gateway Routing Protocol (IGRP), Open Shortest Path First (OSPF) and Intermediate System to Intermediate System (IS-IS)

Exterior Gateway Protocol (EGP) is a Routing Protocol which is used to find network path information between different Autonomous Systems. Exterior Gateway Protocol (EGP) is commonly used in the Internet to exchange routing table information. There is only one Exterior Gateway Protocol (EGP) exists now and it is Border Gateway Protocol (BGP)

Routing Information Protocol (RIP), RIPv1, RIPv2

IP RIP (Routing Information Protocol) comes in two different versions: 1 and 2. Version 1 is a distance vector protocol (RFC 1058) and Version 2 is a hybrid protocol (RFCs 1721 and 1722).

➤ Routing Information Protocol Version 1 (RIPv1)

RIPv1 uses local broadcasts to share routing information. These updates are periodic in nature, occurring, by default, every 30 seconds. To prevent packets from circling around a loop forever, both versions of RIP solve counting to infinity by placing a hop count limit of 15 hops on packets. Any packet that reaches the sixteenth hop will be dropped. RIPv1 is a classful protocol. RIP supports up to six equal-cost paths to a single destination. Equal-cost paths are the paths where the metric is same (Hop count).

➤ Routing Information Protocol (RIPv2)

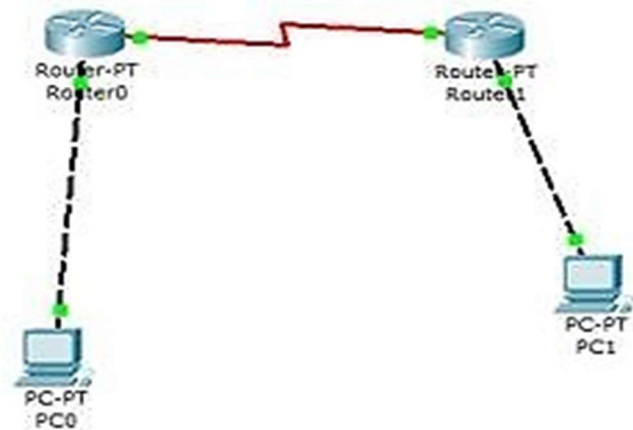
RIPv2 is a distance vector routing protocol with routing enhancements built into it, and it is based on RIPv1. Therefore, it is commonly called hybrid routing protocol.

RIPv2 uses **multicasts** instead of **broadcasts**. RIPv2 supports triggered updates. When a change occurs, a RIPv2 router will immediately propagate its routing information to its connected neighbors. RIPv2 is a classless protocol, and it supports variable-length subnet masking (VLSM).

Both RIPv1 and RIPv2 use hop count as the metric.

Configuration of Routing Information Protocol:

Create topology as illustrated below:

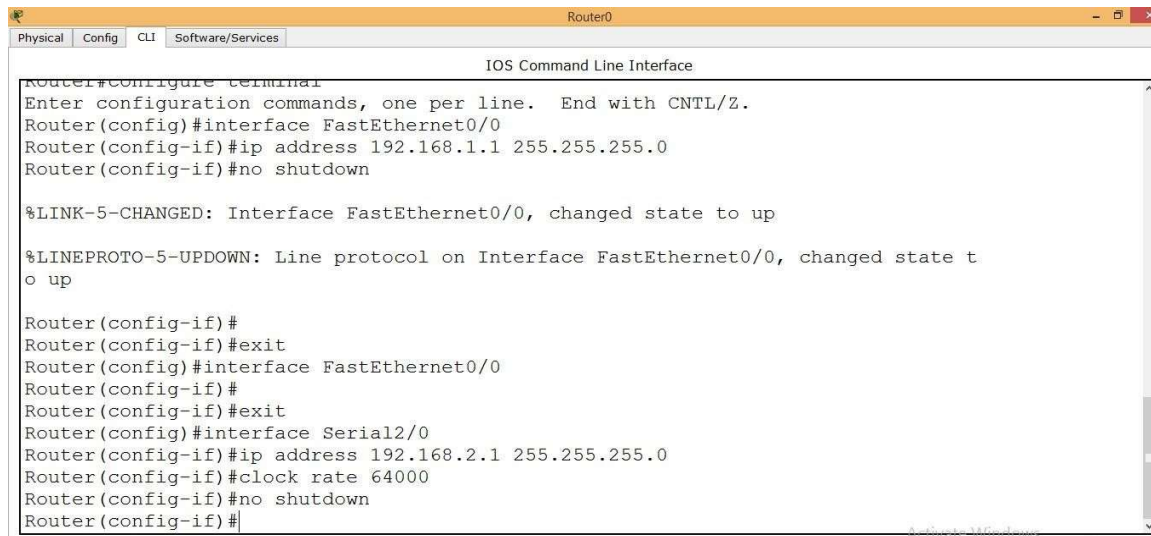


Now, follow the steps as mentioned i.e.

1. We will assign IP addresses to all the fast Ethernet and serial interfaces respectively.
2. We will change the state of the interfaces from down to UP.
3. Then, after we are done with the basic step. We will apply RIP protocol commands on both routers.

IP Configurations of Router 0

Configuring both serial and fast Ethernet interfaces:

A screenshot of the Router0 CLI interface. The window title is "Router0". The tabs at the top are "Physical", "Config", "CLI", and "Software/Services". The "CLI" tab is active, showing the "IOS Command Line Interface". The terminal text shows the configuration of FastEthernet0/0 and Serial2/0 interfaces. The FastEthernet0/0 interface is configured with IP address 192.168.1.1 and 255.255.255.0, and its state is changed to up. The Serial2/0 interface is configured with IP address 192.168.2.1 and 255.255.255.0, and its clock rate is set to 64000. The configuration is completed with the "exit" command.

```
Router0#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

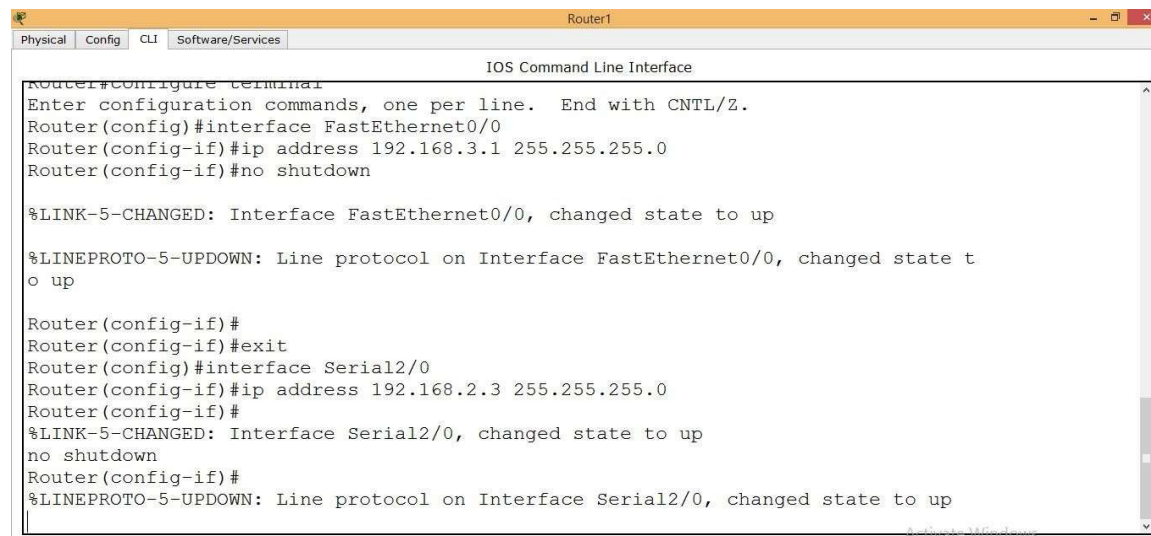
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#
```

IP Configurations of Router 1

Configuring both serial and fast Ethernet interfaces:

A screenshot of the Router1 CLI interface. The window title is "Router1". The tabs at the top are "Physical", "Config", "CLI", and "Software/Services". The "CLI" tab is active, showing the "IOS Command Line Interface". The terminal text shows the configuration of FastEthernet0/0 and Serial2/0 interfaces. The FastEthernet0/0 interface is configured with IP address 192.168.3.1 and 255.255.255.0, and its state is changed to up. The Serial2/0 interface is configured with IP address 192.168.2.3 and 255.255.255.0, and its state is changed to up. The configuration is completed with the "exit" command.

```
Router1#configure terminal
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface FastEthernet0/0
Router(config-if)#ip address 192.168.3.1 255.255.255.0
Router(config-if)#no shutdown

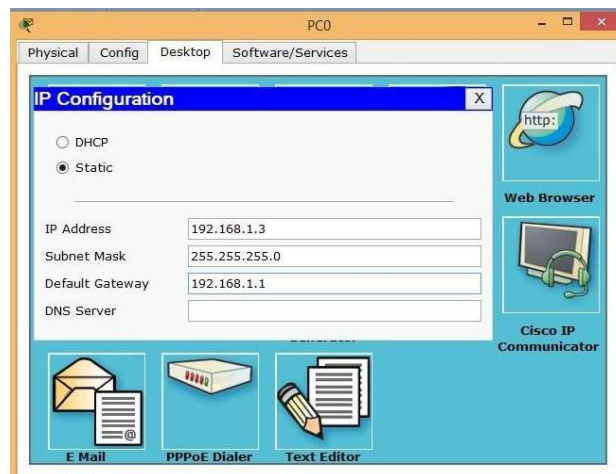
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.168.2.3 255.255.255.0
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
no shutdown
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up

Router(config-if)#
```

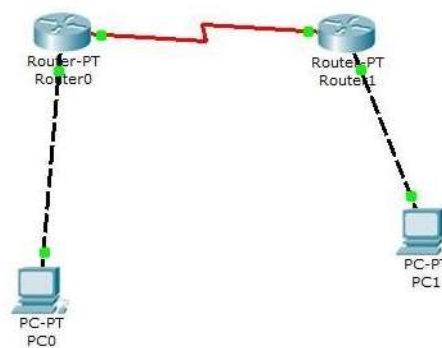
Assigning IP address to PC0



Assigning IP address to PC1



After Ip configuration and changing links state to up this is how our topology will look like:



Configuration of RIP Protocol

R0 Configuration

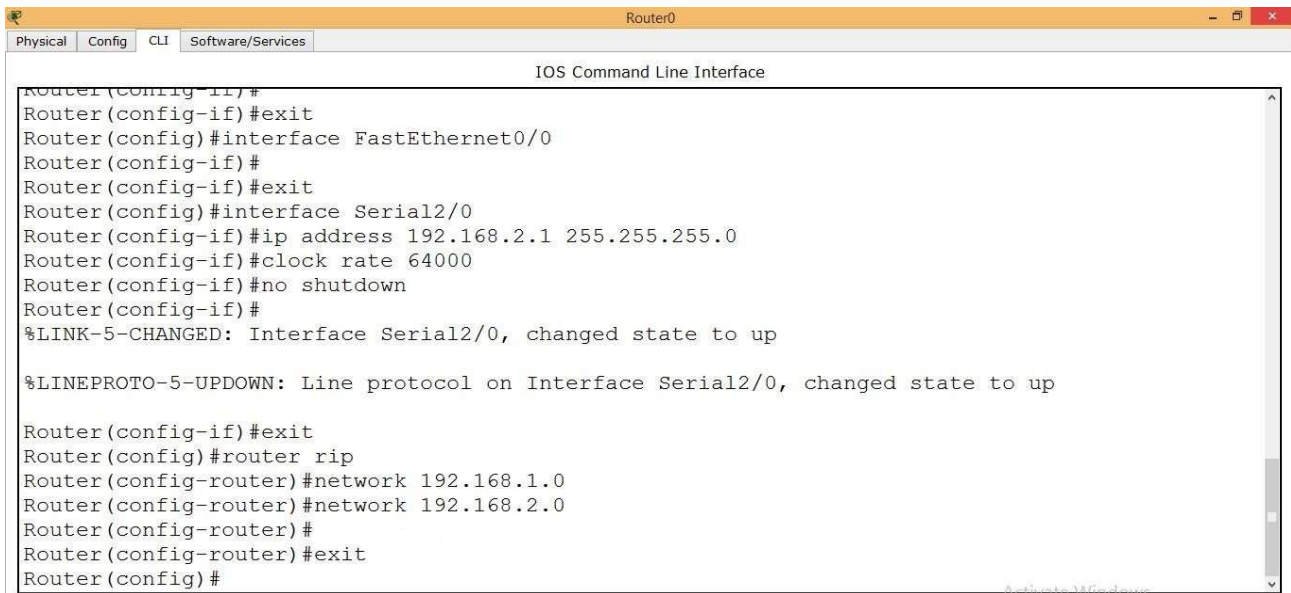
To apply protocol RIP, we will write the following set of commands.

```
Router(config)# router rip
```

```
Router(config-router)# network 192.168.1.0
```

```
Router(config-router)# network 192.168.2.0
```

```
Router(config-router)#exit
```



```
Router0
Physical Config CLI Software/Services
IOS Command Line Interface
Router(config-if)#
Router(config-if)#exit
Router(config)#interface FastEthernet0/0
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0, changed state to up
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#network 192.168.1.0
Router(config-router)#network 192.168.2.0
Router(config-router)#
Router(config-router)#exit
Router(config)#
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```

R1 Configuration

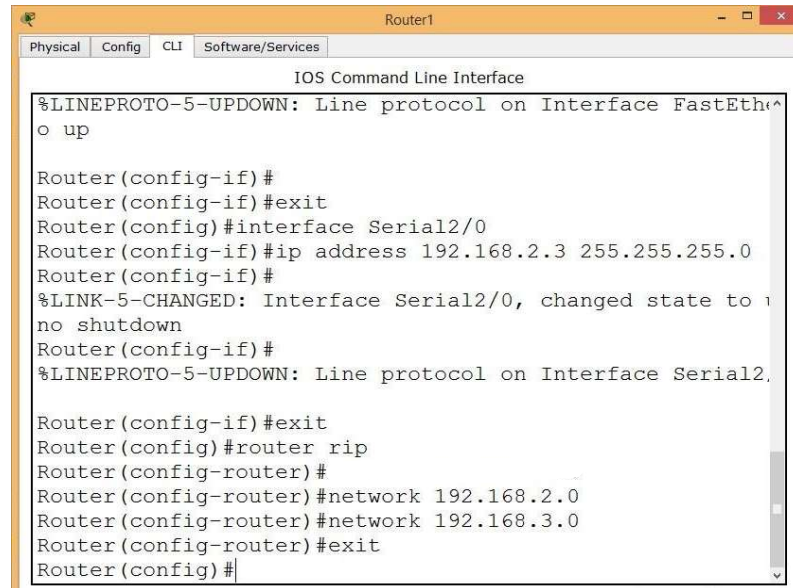
To apply protocol RIP, we will write the following set of commands on R2 as well.

```
Router(config)# router rip
```

```
Router(config-router)# network 192.168.2.0
```

```
Router(config-router)# network 192.168.3.0
```

```
Router(config-router)#exit
```



```
Router1
Physical Config CLI Software/Services
IOS Command Line Interface
%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0 is now up
Router(config-if)#
Router(config-if)#exit
Router(config)#interface Serial2/0
Router(config-if)#ip address 192.168.2.3 255.255.255.0
Router(config-if)#
%LINK-5-CHANGED: Interface Serial2/0, changed state to up
no shutdown
Router(config-if)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial2/0 is now up
Router(config-if)#exit
Router(config)#router rip
Router(config-router)#
Router(config-router)#network 192.168.2.0
Router(config-router)#network 192.168.3.0
Router(config-router)#exit
Router(config)#
```

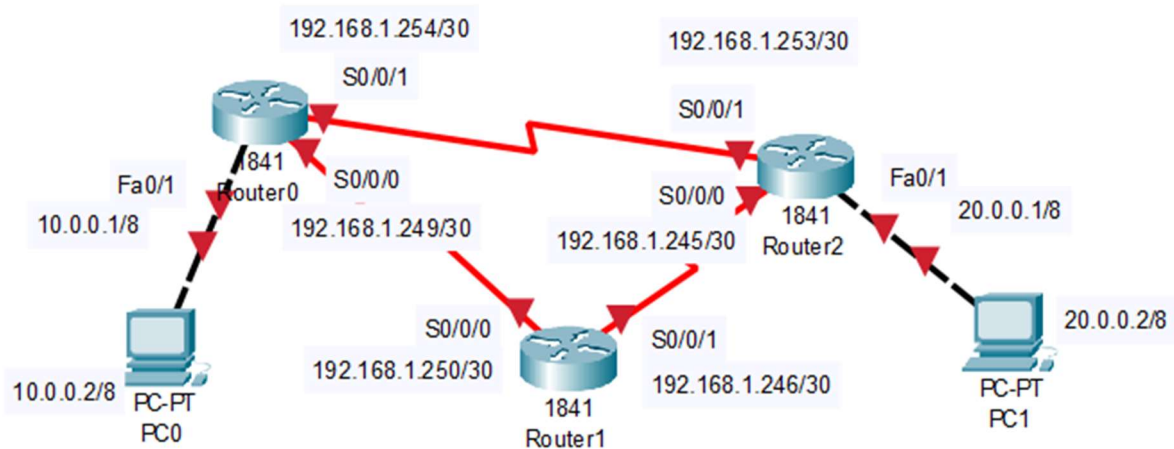
Write all the commands shown in the above steps. Now, check if the Traffic is enabled and you can easily send data from PC0 to PC1.

Lab Task

Create the following topology and simulate a configuration Routing Information protocol.

- Implement RIP version 2.
- Ping from PC0 $\leftarrow \rightarrow$ PC1. Is the ping Successful? Also Trace route from PC0 $\leftarrow \rightarrow$ PC1.
- Shut down link b/w Router 0 and Router 1.
- Again, Trace route from PC0 $\leftarrow \rightarrow$ PC1. Mention changes in the routing path.

Write descriptive output as well as attach results of all the above stated tasks.



Conclusion
